

ASSIGNMENT - I

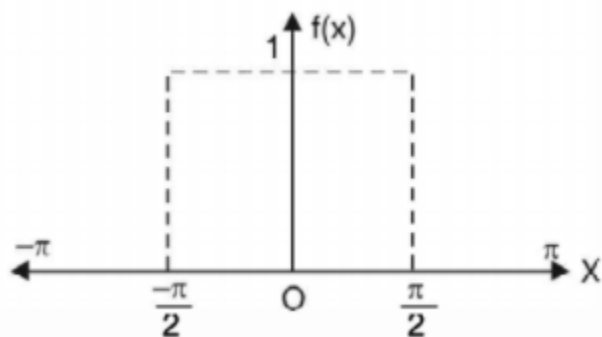
Introduction to Fourier Series, Dirchlet's conditions, Euler's formulae:

1. Obtain the Fourier coefficient a_0 using Euler's formula for the periodic function

$$f(x) = \begin{cases} x & -\pi < x < 0 \\ -x & 0 < x < \pi \end{cases}$$

Ans : $a_0 = -\pi$

2. Obtain the Fourier coefficient a_0 using Euler's formula for the periodic function



Ans : $a_0 = 1$

3. Obtain the Fourier coefficient b_1 using Euler's formula for the periodic function

$$I = \begin{cases} I_0 \sin \theta & \text{for } 0 < \theta \leq \pi \\ 0 & \text{for } \pi < \theta \leq 2\pi \end{cases}$$

Ans: $b_1 = \frac{I_0}{2}$